

Vish Karthikeyan

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Location: San Francisco Bay Area

EDUCATION

Stanford University, School of Engineering, California

Bachelor of Science Degree in Computer Science (Theory track)

Minor in Public Policy

Expected June 2027

Activities and Societies: Director of Outreach: Stanford Political Union • Counselor: Bridge Peer Counseling Center • Teacher: Stanford Splash • Writer: Stanford Political Review • Mentor: Bechtel International Center

Tentative Master of Science Degree in Computer Science (Artificial Intelligence track)

Expected June 2028

CS Academy Erode, India

All India Senior Secondary Certificate, Top percentile national distinction with 96% aggregate (100% in Computer Science)

May 2023

All India Secondary School Exam, Top percentile national distinction with 98% aggregate

May 2021

Activities and Societies: School Captain • National Debate Finalist • National Talent Search Examination Scholar

RELEVANT COURSEWORK

CS 229 (Machine Learning; graduate-level)

CS 124 (Natural Language Processing)

CS 161 (Design and Analysis of Algorithms)

CS 254 (Complexity Theory; graduate-level)

CS 106B (Abstract Data Structures)

MATH 104 (Applied Linear Algebra)

ECON 50Q and 51 (Quantitative Microeconomic Analysis)

CS 109 (Probability for Computer Scientists)

SELECTED PROJECTS

Bias Evaluation in LLM vs. Human Criminal Judgments (Ongoing, 2025–Present): Conducted matched-guise experiments comparing commercial LLMs (e.g., GPT-series, Claude) to human professionals on guilt/sentencing decisions using controlled case vignettes; analyzed linguistic cues for bias divergence in probabilistic vs. biological reasoning with Hugging Face Transformers and statistical metrics.

Reducing Hallucinations in LLM Coding Agents via Uncertainty-Aware Decisions (2026; CS 229 Final Project): Framed tool-calling as uncertainty-guided decision-making; trained agents with supervised fine-tuning and reinforcement learning to defer requests instead of hallucinating invalid APIs/logic, reducing errors in programming tasks using PyTorch, Hugging Face, and uncertainty signals (e.g., entropy).

Provable Expressivity Limits of Transformer Architectures (2026; CS 254 Final Project): Analyzed circuit complexity and language recognition capabilities of transformer encoders/decoders under varying attention mechanisms (hard/soft, unique/averaged) and depths; derived impossibility results via diagonalization and learning theory bounds to explain empirical vs. theoretical performance gaps.

WORK EXPERIENCE

Student Dining Worker, Stanford Residential and Dining Enterprises

Fall 2025 - current

- Assisted at Stanford's largest dining hall, serving 3000+ people a day in an extremely fast-paced, high-volume environment with hands-on experience in food service operations
- Helped bridge communication and rapport between the student body and dining staff, contributing to a more responsive and student-centered campus dining experience

Residential Assistant, Summer Session, Stanford University

Summer 2025

- Built an inclusive, supportive residential community for 300+ high school summer students
- Planned and led 3+ programs a week fostering connection and well-being
- Served as a key resource for student concerns and campus referrals; coordinated case management

Teaching Assistant, Frosh 101, Stanford University

Fall 2024

- Inspired community-building conversations and activities amongst a cohort of freshmen, shaping their college transitions
- Attended a quarter-long course on inclusive teaching practices and research on the impact of mental health, diversity, inclusion and sense of belonging on learning

Team Leader, Reunion Homecoming, Stanford Alumni Association

Fall 2024

- Led planning efforts, coordinated event logistics, and developed programming for alumni
- Recruited, trained, and managed a team of volunteers, delegating tasks and providing guidance to ensure efficient event execution

Neighborhood Council Representative, Stanford Residential Education

2023-24

- Managed \$200,000 in funding across community building events for 800+ students, engineering various types of events and festivals
- Designed permanent community spaces for the Stanford Outdoor Enhancement Project

Associated Students of Stanford University

2023-24

- Assistant Elections Commissioner:** Facilitated student government elections involving 18,000 voters, supervised data management and security and practiced constitutional hermeneutics and policy development
- Senate Associate:** Participated in a leadership and advocacy workshop for six months, directed a team working on designing and implementing modifications to Stanford Dining

SKILLS AND INTERESTS

Programming

Python (including SciKit-Learn, PyTorch, TensorFlow, NumPy Pandas and Hugging Face), C/C++

Technical

Git and version control, UNIX command-line, SQL (MySQL, SQLite), CSS, HTML, Javascript, RESTful APIs, Docker and containerization, Kubernetes, Agile development, Jupyter, supervised and unsupervised learning, data structures, algorithmic complexity, model evaluation and metrics, feature engineering and data preprocessing

Langauges

English (native proficiency), Tamil (native proficiency) and Spanish (elementary)

Interests

Natural language processing and transformers, large language models, computability and complexity theory, causal inference for public policy, political theory, mental health and peer counseling